MASONRY

Brick Work In Cement Mortar

- Masonry may be defined as the construction of building bounded together with mortar.
- Masonry is normally used for construction of walls.

It performs a variety of functions, such as

- Supporting loads
- Subdividing space
- Providing thermal and acoustic insulation
- Affording fire and weather protection etc.



masonry



mortar

Mortar

- It is the pasty material formed by the addition of water mixture composed of fine aggregate (sand) and cement or lime.

- Generally the following types of mortar are in use:

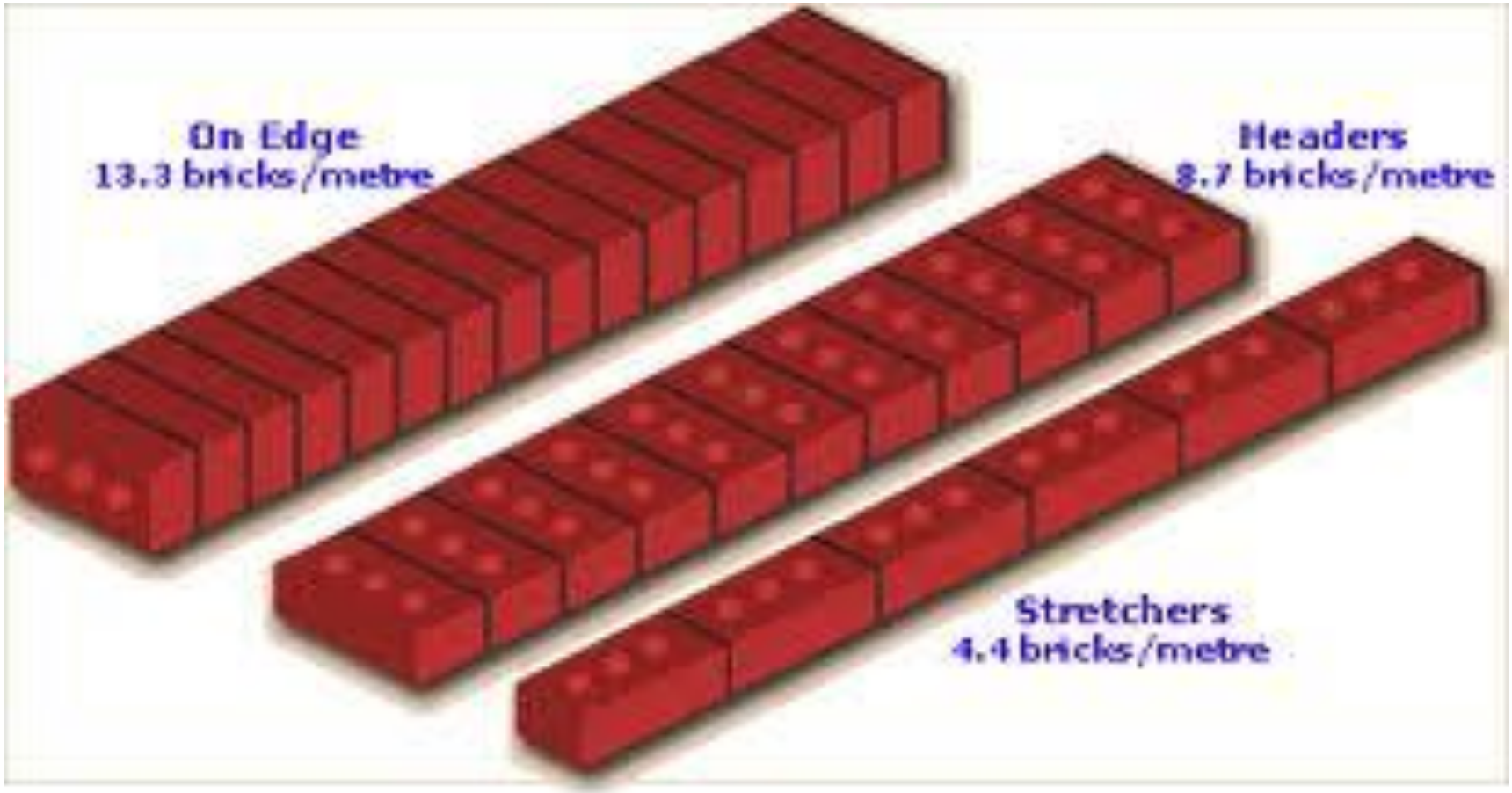
1. Mud mortar
2. Lime mortar
3. Lime surkhi mortar
4. Cement mortar and
5. Cement lime mortar or lime- cement- mortar

Terms Used in Brick Masonry

- ***Course:*** A complete layer of bricks laid on the same called a course and its thickness is equal to the thickness of one mortar joint.
- ***Frog :*** Depressions provided in the face of the brick.
- ***Bed:*** The bottom surface of the brick when it is laid flat is called bed.
- ***Stretcher:*** The side surface of bricks visible in elevation when the brick is laid flat is called the stretcher (19 cm x 9 cm).
- ***Header:*** The end face of the brick when it is laid flat is as header (9cm x 9cm)

Terms Used in Brick Masonry

- ***Bed joint:*** The horizontal mortar joints between two successive courses are known as bed joints.
- ***Perpend*:** These are the joints between bricks either in longitudinal or cross directions.
- ***Bat:*** The brick which is cut across its width is called as bat
- ***Closer:*** The brick which is cut across its length is called as closer.


On Edge
13.3 bricks/metre
Headers
8.7 bricks/metre
Stretchers
4.4 bricks/metre

Principles in Brick Masonry Construction

- Good brick masonry should utilize bricks which are sound, hard, well burnt, and tough with uniform colour, shape and size.
- The brick should be compact, homogeneous and free from holes, cracks, air-bubbles and stone lumps.
- In the brick work, the bricks should be laid on their beds with the frogs pointing upwards.
- The brick courses should be laid truly horizontal and should have truly vertical, joints.
- As far as possible the use of brick-bats should be discouraged.

- As far as possible the brick walls should be raised uniformly with proper bond. Generally the height of brick masonry construction in a day should be less than l.5 m.
- In order to ensure continuous bond between the old and the new, the walls should be stopped with a. toothed end.
- Finished brickwork in lime mortar should be cured for a period of 2 to 3 weeks. This period can be reduced to 1 to 2 weeks in case of brickwork with cement mortar.
- In order to carry out the brickwork at higher level single scaffolding is used.

Bonds in Brick Masonry

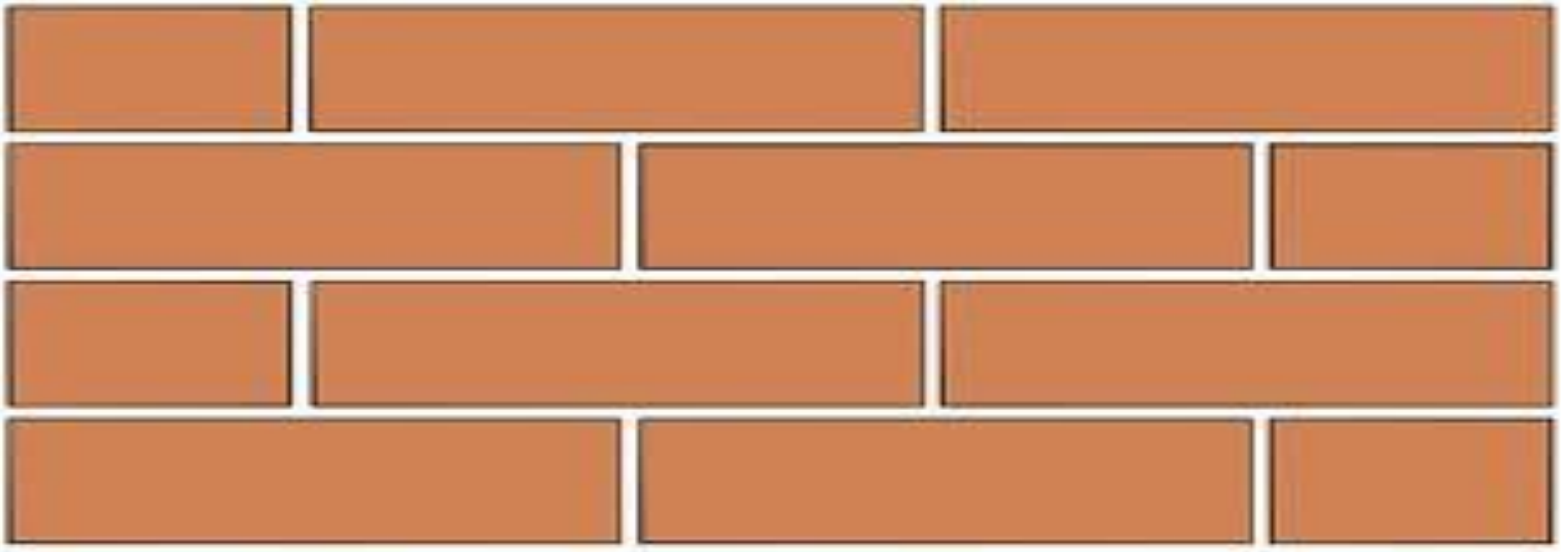
- Bonding is a process of arrangement of bricks and mortars to tie them together in a mass of brickwork.
- It should have a minimum of vertical joints in any part of the work. It is a weak portion of brickwork and should not be continuous in two successive courses.
- A wall having continuous vertical joints shall act as independent column. Hence load on wall shall not be uniformly distributed.
- A more strong and durable brick masonry construction will not have continuous vertical joints. It shall distribute load on a wider area and thereby minimise the tendency to settle.

various types of bonds are:

- Stretcher bond
- Header bond
- English bond
- Flemish bond

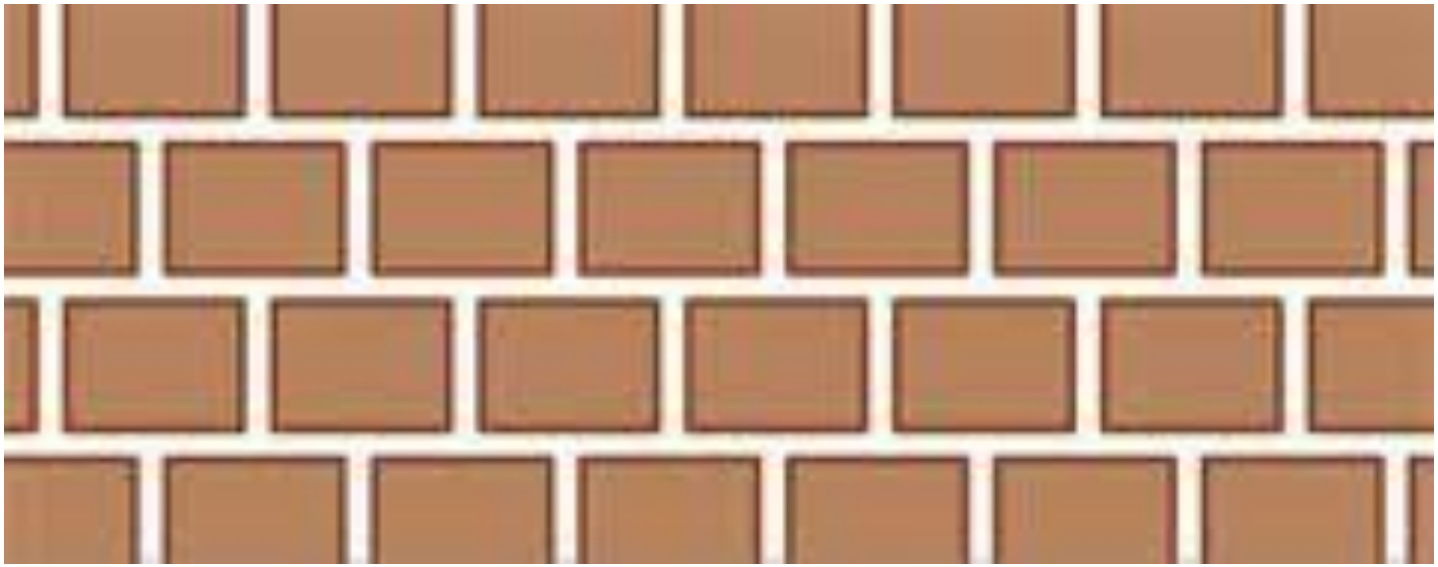
1. Stretcher bond

- In this type of bond all the bricks are laid with their lengths in the direction of the wall.
- This pattern is used for walls having thickness of 9 cm only.



2. Header bond

- In this type of bond all the bricks are laid with their ends towards the face of the wall.
- This type of arrangement is suitable for walls, which are one brick thick.



3. English bond

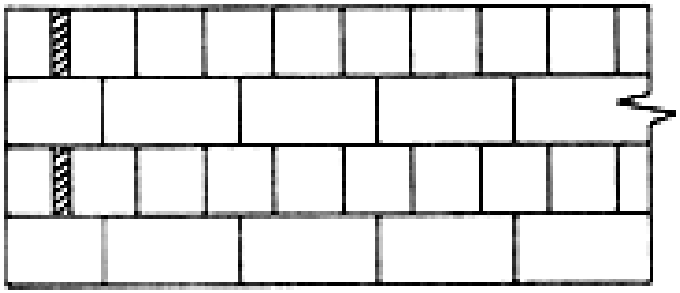
- In this type of bond alternate courses of headers and stretchers are laid.
- It is necessary to place the queen closer after first header in the heading course for breaking the joints vertically.

- A quoin closer must be provided after the quoin header. A header course should never start with a quoin closer.
- Each alternate header should be centrally placed over a stretcher.
- Continuous vertical joint should not be allowed except at stopped end.

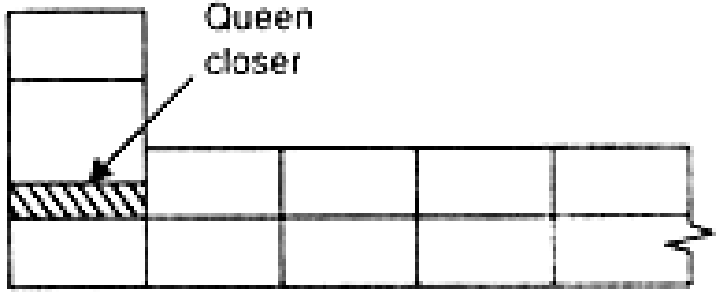
English bond contd…..

- In case of the wall thickness equivalent to an even number of half bricks the wall shall present similar appearance of both the faces.
- In case of the wall thickness equivalent to an odd number of, half bricks, the same courses shall have stretchers on one and header on the other.
- The joints on the header course should be made thinner than those in the stretcher course. This is because of the fact that the number of vertical joints in the stretcher course are half the number of joints in the header course.

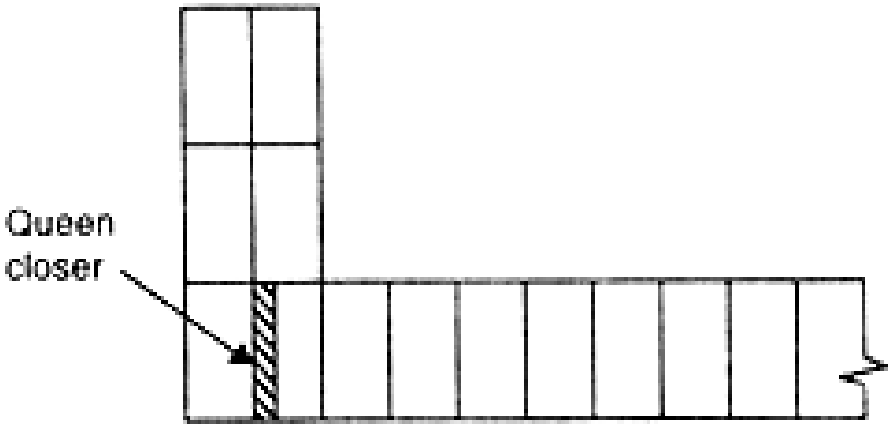
English bond



Elevation-English bond

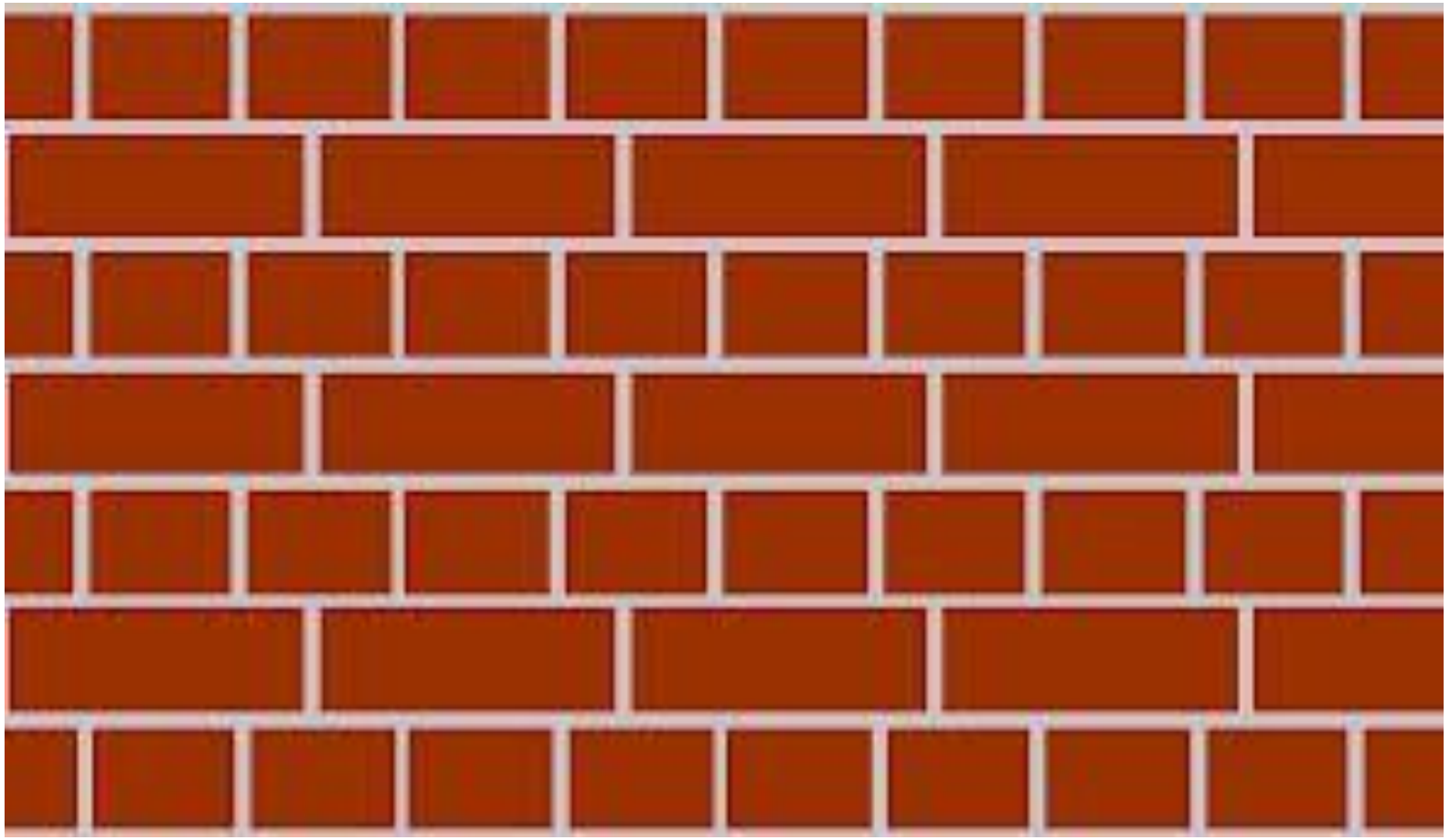


Plan of stretcher course



Plan of header course

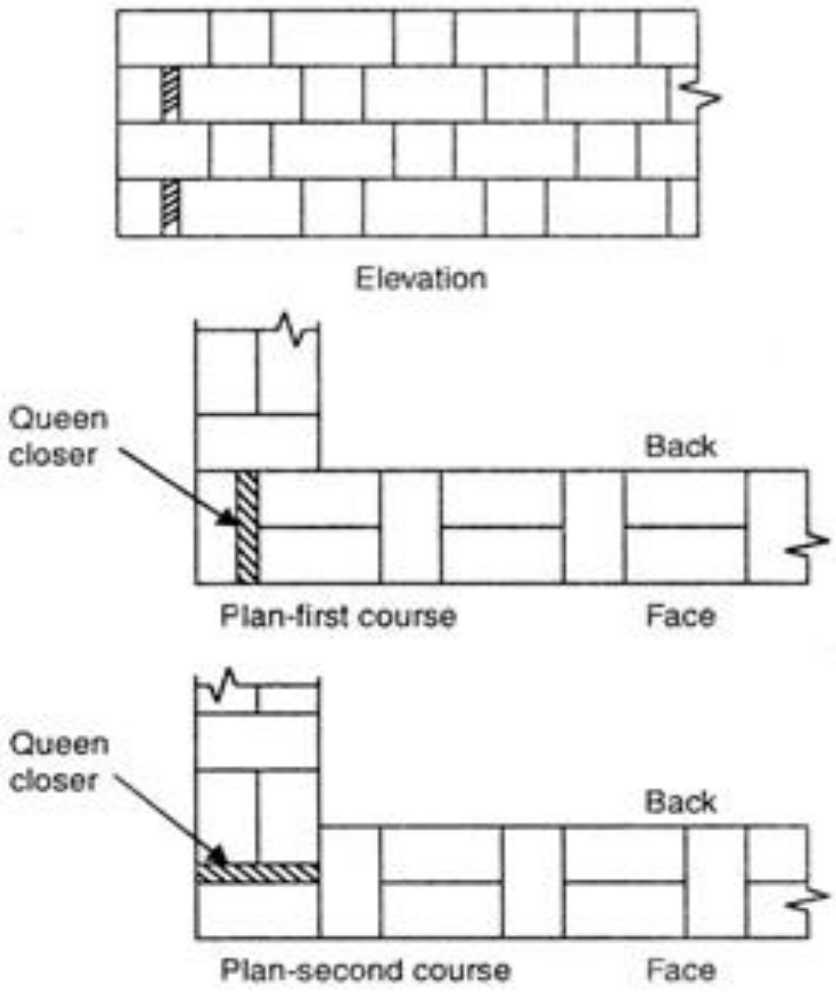
English bond



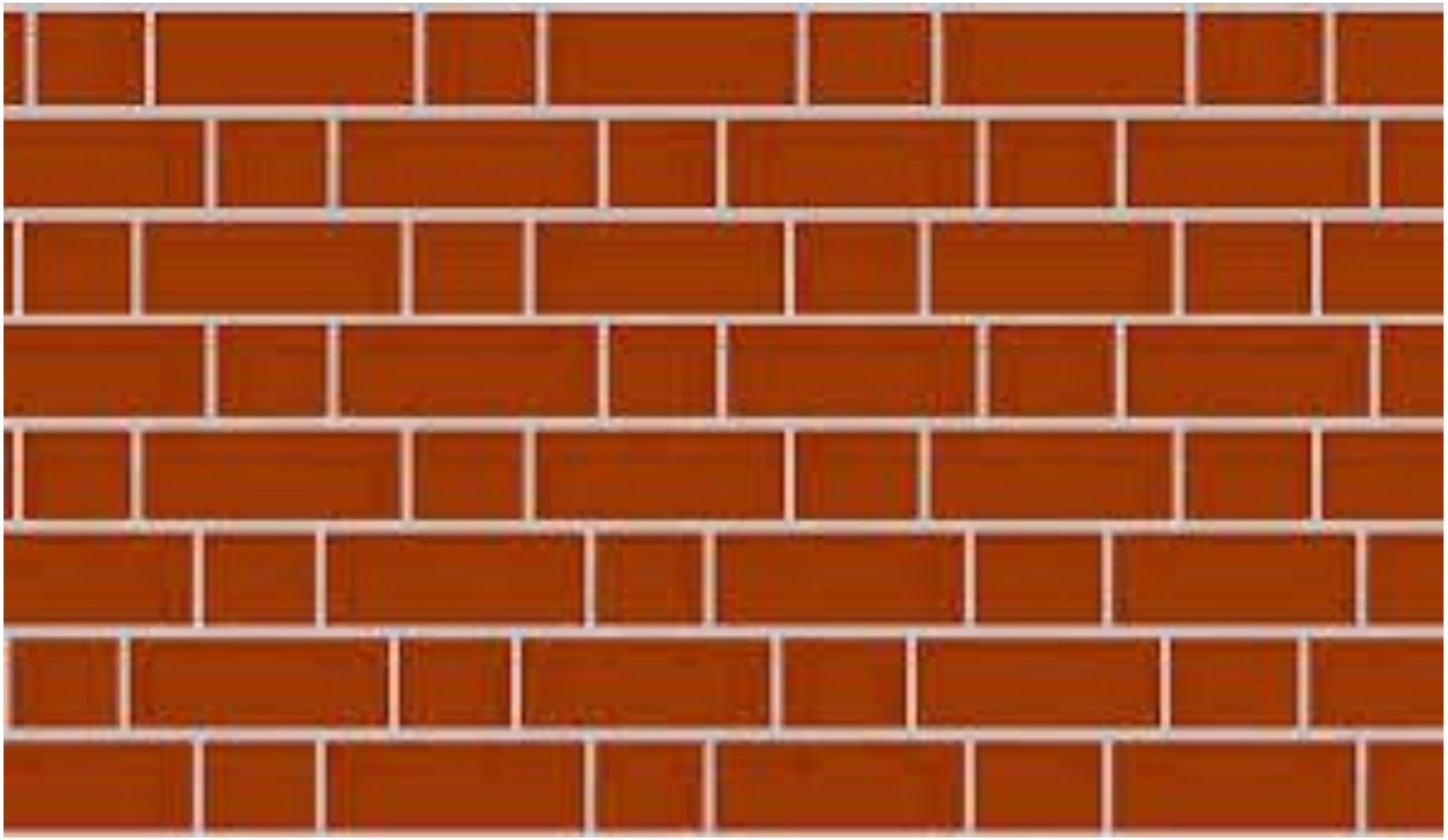
4. Double Flemish bond

- In this bond alternate headers and stretchers are laid in each course.
- This type of bond is better in appearance than the English bond. The facing and back are of the same appearance.
- The Brick bats are used in case of walls having thickness equivalent to odd number of half bricks.
- The queen closer is placed next to the quoin header in alternate, courses in order to break the continuity of the vertical joints.

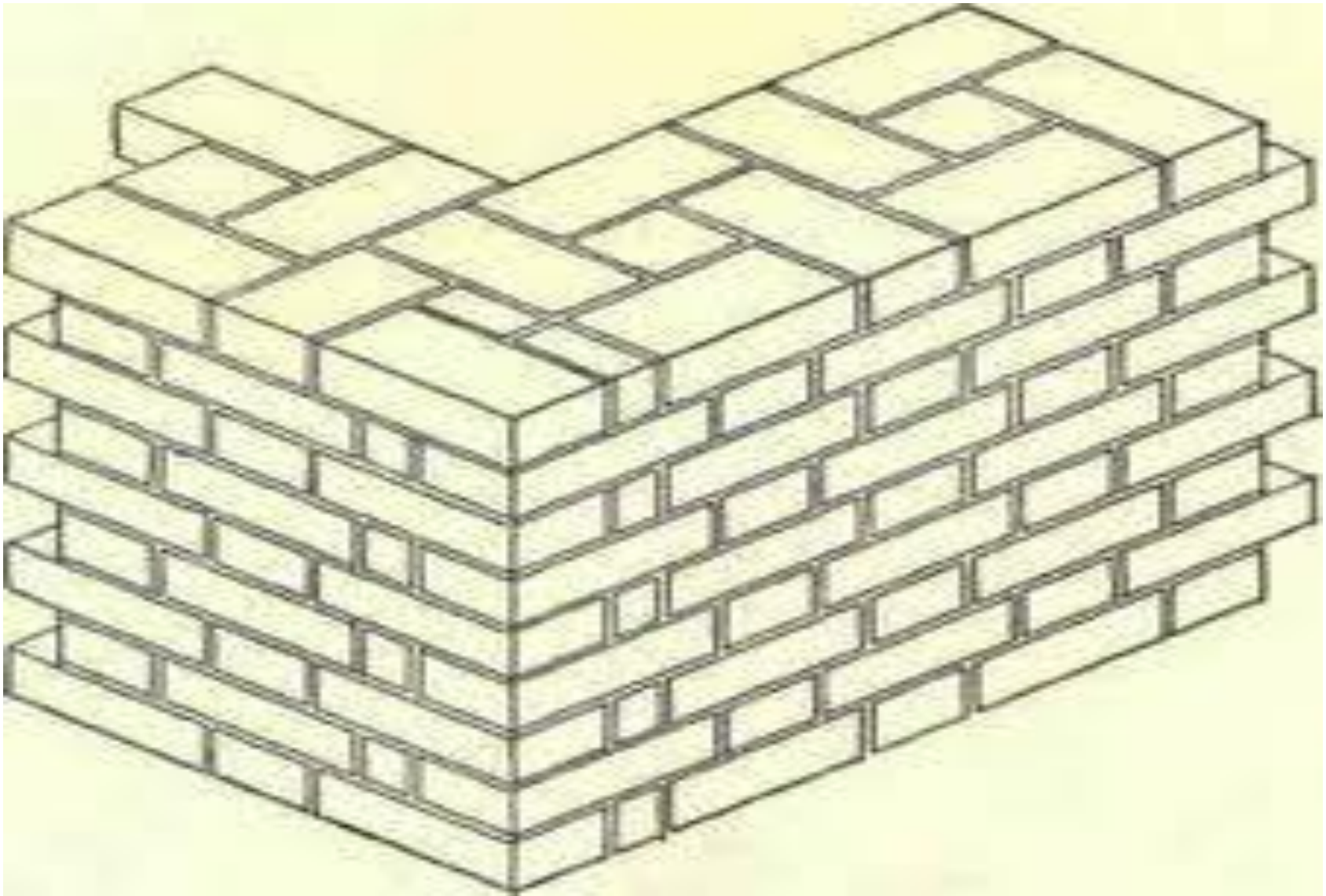
Double Flemish bond



Flemish bond



Double flemish bond



STONE MASONRY

Classification of Stone Masonry

Stone Masonry

Rubble Masonry

1. Random rubble masonry
   a) Coarsed
   b) Uncoarsed
2. Square rubble masonry
   a) Coarsed
   b) Uncoarsed
3. Polygonal rubble masonry
4. Flint rubble masonry
5. Dry rubble masonry

Ashlar Masonry

1. Ashlar fine
2. Ashlar rough tooled
3. Ashlar rock or quarry faced
4. Ashlar chamfered
5. Ashlar block in course

RUBBLE MASONRY

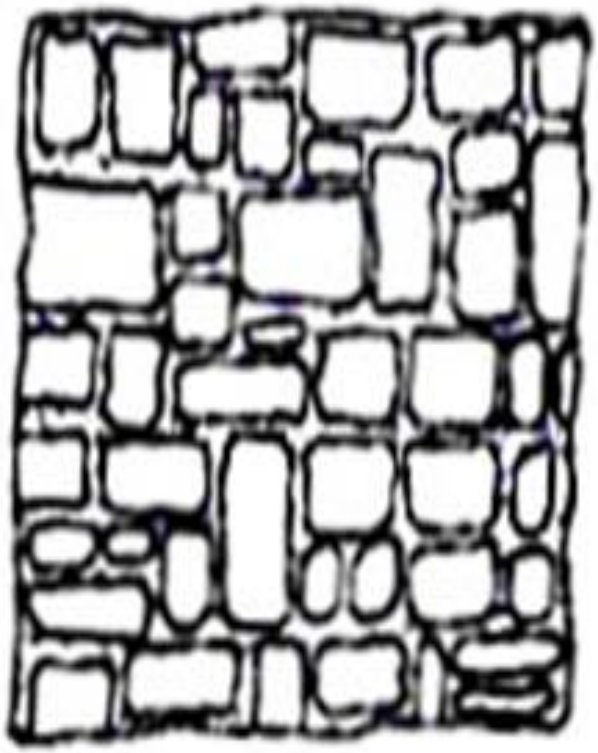
- In this type of masonry, stones of irregular sizes and shapes are used.
- The stones, as obtained from quarry, are taken for use in the same form or they are broken and shaped in suitable sizes by means of hammer as the work proceeds.

1.Random rubble masonry:

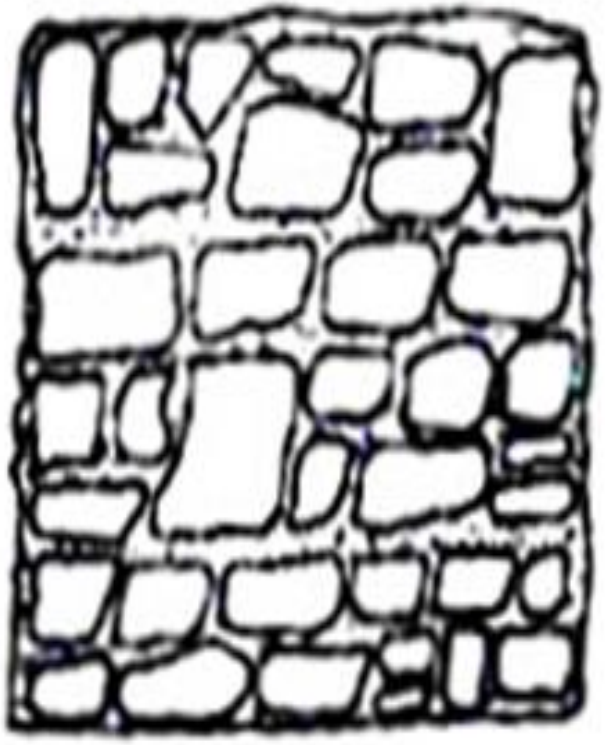
- In this type of masonry, the stones used are of widely different sizes. This is the roughest and cheapest form of stone masonry.
- In coarsed random rubble masonry, the masonry work is carried out in courses such that the stones in a particular course are of equal heights.

Rubble masonry





Uncoarsed Random Rubble Masonry



Coarsed Random Rubble Masonry

2.Square rubble masonry

- In this type of masonry stones having straight bed and sides are used.
- The stones are usually squared and brought to a hammer dressed or straight cut finish.
- In the coursed squared rubble masonry, the work is carried out in courses of varying depth.

3.Dry rubble masonry

- In this type of masonry mortar is not used in the joints. This type of construction is the cheapest and requires more skill in construction.
- This may be used for non-load bearing walls such as compound wall etc.

Square random rubble masonry

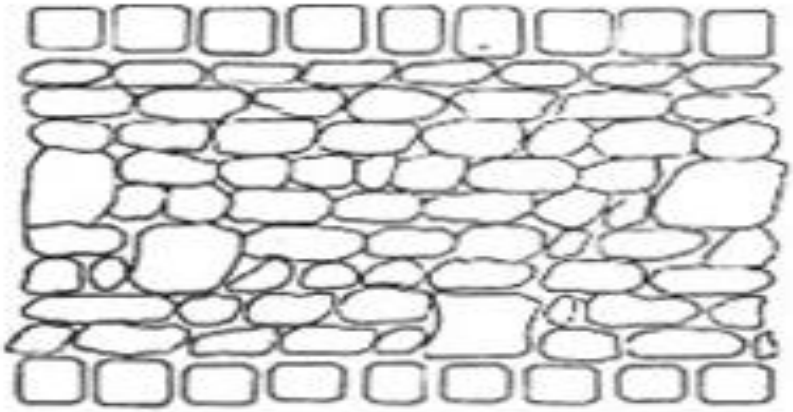


Dry rubble masonry



4.Flint rubble masonry

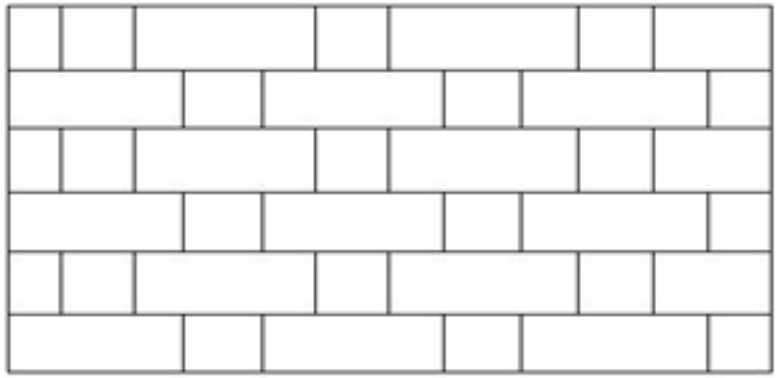
- In this type of rubble masonry, stones used are flints or cobbles.
- These are irregularly shaped modules of silica. The stones are extremely hard.



Flint Rubble Masonry

ASHLAR MASONRY

- In this type of construction, no irregular stones are used. The stones used in this masonry are rectangular blocks and are all dressed finely with chisel.
- The courses are not necessarily of the same height. It may vary from 25 to 30 cm.



Ashlar masonry



Different types of ashlar masonry are

❖ **Ashlar rough tooled masonry**

- In this type of ashlar masonry the beds and sides are finely chisel-dressed.
- But the face is made rough by means of tools.

❖ **Ashlar rock or quarry faced masonry**

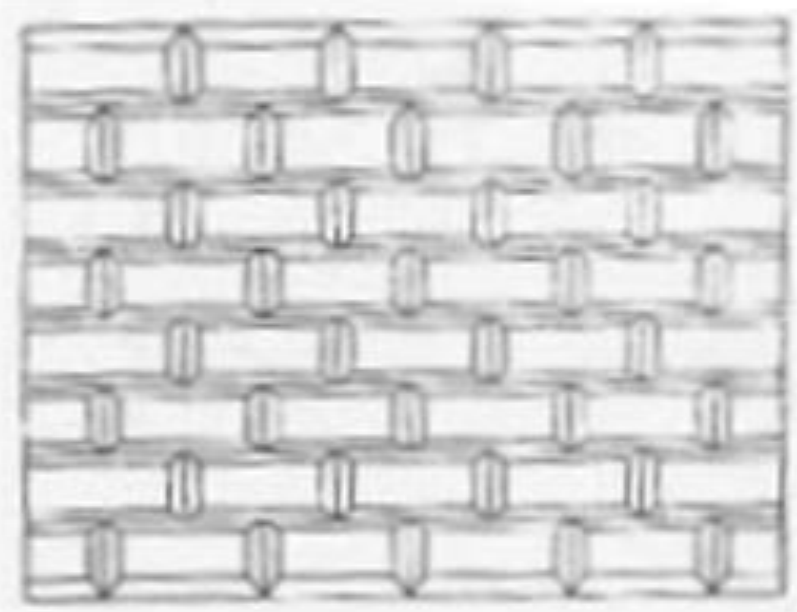
- In this type of ashlar masonry, a strip about 25 mm wide and made by means of chisel is provided around the perimeter of every stones as in case of rough-tooled ashlar masonry.

❖ Ashlar chamfered masonry

- In this type of ashlar masonry, the strip is provided as above. But it is chamfered or bevelled at an angle of 45 degrees by means of a chisel for a depth of about 25 mm.

❖ Ashlar block-in-coarse masonry

- This is combination of rubble masonry and Ashlar masonry.
- In this type of masonry, the face work is provided with rough tooled or hammer dressed stones and backing of the wall may made in rubble masonry.



Ashlar Chamfered Masonry

Points to be observed in the construction of stone masonry

- Stones should be strong, rough and hard
- Each stone should be well watered before use
- All the stones should be laid on their natural bed
- Proper bond should be maintained; formation of vertical joints should be avoided.
- Small stone pieces should be used for facing.
- The wall should be raised uniformly throughout its length.
- Stones should be dressed properly according to the type of masonry.
- Mortar should be in proper proportion.
- After the construction is over the whole masonry work should be kept wet for at least 2 to 4 weeks.

Comparison between Stone Masonry and Brick Masonry

- Cost of stone masonry work is more, as it requires more skilled labour.
- Due to various sizes and shapes of stone, complicated lifting devices are required in the construction. But bricks having regular shape and uniform size, can be moved easily by manual labour.
- In the case of brick masonry any mortar can be used, but in the case of stone masonry mortars other than cement will not be having any bond with the stone surface.
- In the case of stone masonry, the dead weight is more because it is comparatively heavy, for the same reason, it is suitable for under water construction.
- Stone work is stronger than brick work.

- Thinner walls are not possible in stone masonry.
- Bricks resist fire better than stones. In case of fire, stone easily disintegrates.
- In the case of brick masonry, mortar joints are thin and uniform due to their uniform size and shape, and hence the structure becomes more durable with less consumption of mortar. But in the case of stone work, - pointing is necessary.
- It is easy to provide connections and openings in case of brick work.
- Brick work is less water tight than stone work, bricks absorb moisture from atmosphere and dampness can enter the building. Hence plastering is essential, as extra expenditure to brick work